



TDA6323 Algorithm Design and Analysis

PROJECT GUIDELINES

This group project contributes to the **final assessment** and carries a total of **30%** of the final grade.

1) Group Formation

- Form a group of **at most FOUR (4) members from the same lab section only**.
- **No inactive ("sleeping") members are allowed.** If a group member is not contributing, report it immediately. The reported member will be required to complete and submit the project individually.
- If an inactive member is reported on the final submission, they will receive zero marks unless they can provide proof of their contribution.

2) Topic Selection

- You may **select a topic from the given list below or propose your own**. If a topic is too general, you should narrow it down and focus on a specific subtopic.
 - *Example:* If choosing "Sorting Data in Arrays," instead of covering all sorting techniques, focus on one, such as Quick Sort, which has multiple improved versions for comparison.
 - *Example:* If choosing "Graph Algorithms," compare two different algorithms that solve similar problems, such as Prim's vs. Kruskal's algorithms for finding a Minimum Spanning Tree in network communication design
- **Topic Registration:** Register your Group member and topic using the [TDA6323 Group Registration Form.xlsx](#)
- **NO duplicated topics** are allowed **within the same lab session**.
- **Registration Deadline:** **14/04/2025 by 5pm (Week 3)**

Table 1: List of Topics

No	Application
1	Sudoku Puzzle
2	Job Assignment
3	Shortest Path/Route/Network
4	Search Engine focusing on Web Crawlers
5	Crossword puzzle
6	Chess board
7	Knapsack

8	Task Scheduling
9	Sorting Images/Objects
10	Randomization in Machine Learning
11	Travelling Salesman Problem
12	Feature Selection
13	Clustering (suggestion: choose one clustering technique but different algorithm design approach)
14	Classification (suggestion: choose one classifier but different algorithm design approach)
15	Indoor Positioning
16	Resource Management
17	Image Segmentation
18	Text Segmentation
19	Web Searching
20	Recommendation System

3) Project Requirements

This project consists of **TWO main parts**:

Part 1: Literature Review and Research

a. Introduction & Research Questions

Provide a clear description of the selected application, including:

- What the application is.
- What problems it solves.
- The initial version of the algorithm used.
- Improvements made by researchers over time.

Define **3 to 4 research questions** to guide your literature review. Example questions:

- What improvements have been made to the algorithm?
- How do the algorithm designs differ from one another?
- What are the strengths and weaknesses of each design?
- What challenges remain?

b. Review of Different Algorithm Designs

Choose one of the following approaches:

1. Review **at least THREE (3)** improved versions of a similar algorithm. OR
2. Compare **at least THREE (3)** different algorithms that solve the same problem.

Each algorithm review should include:

- The original inventor(s) of the algorithm.
- The design technique used (e.g., dynamic programming, divide and conquer, recursion, linked lists, matrices, etc.).
- The motivation behind the improvement, why the improvement is suggested
- Strengths and weaknesses of each approach.
- Any remaining challenges or limitations.

Include at least **20 research papers** in your literature review and **80% of the references must be from the last five years (2021 – current)** to ensure relevance and up-to-date research.

A sample of good review paper is shared (see Appendix 2).

****Tips to perform search in getting useful articles****

- Go to <https://scholar.google.com.my/>
- Key in the search keywords (refer to Appendix 2 and define a proper sets of keywords), such as 'Quick sort review' (see Figure 1).

The screenshot shows three search results for 'QuickSort' on Google Scholar. Each result includes the title, author, journal name, year, and a brief abstract. The first result is 'Distributional convergence for the number of symbol comparisons used by QuickSort' by J.A. Fillard, published in 'The Annals of Applied Probability' in 2013. The second result is 'A Review on Different Quicksort Algorithms' by SM Deshmukh and AK Bhavsar, published in 'PRATIBHA: INTERNATIONAL JOURNAL' in 2020. The third result is 'Comparative study of two divide and conquer sorting algorithms: quicksort and mergesort' by OE Taiwo and AO Christianah, published in 'Procedia Computer Science' in 2020. Each result has a '[PDF]' link and a 'Cited by' count.

Distributional convergence for the number of symbol comparisons used by **QuickSort** [PDF] projecteuclid.org
 J.A. Fillard - The Annals of Applied Probability, 2013 - projecteuclid.org
 ... of key comparisons for **QuickSort**. In this subsection we **review** four known **QuickSort** key-...
 Smoothness and decay properties of the limiting **Quicksort** density function. In Mathematics ...
 ☆ Save Cite Cited by 11 Related articles All 12 versions

[PDF] A Review on Different Quicksort Algorithms [PDF] sscoetjalgaon.ac.in
 SM Deshmukh, AK Bhavsar - PRATIBHA: INTERNATIONAL JOURNAL ..., 2020 - sscoetjalgaon.ac.in
 ... Based on **quick sort** parallel **quick sort** algorithm has been ... **quick sort** algorithm with its
 complexity and parallel **quick** ... here for analyzation ie **Quick sort** of parallel and Hyper **quick sort** ...
 ☆ Save Cite Related articles All 4 versions

Comparative study of two divide and conquer sorting algorithms: **quicksort** and **mergesort** [PDF] sciencedirect.com
 OE Taiwo, AO Christianah, AN Oluwatobi ... - Procedia Computer ..., 2020 - Elsevier
 ... Such include **Quick sort** and Merge **sort** sorting algorithms. ... sizes, **Quick sort** algorithm is
 faster, though Merge **sort** algorithm is ... case for **quicksort** is $O(n^2)$ and that of merge **sort** remains ...
 ☆ Save Cite Cited by 3 Related articles All 2 versions

Figure 1. Sample Output After Search

- Click the relevant articles. In case if payment is needed, you can try to access by using Open Athens where login through MMU IDM username and password are needed. For databases that have subscribed by MMU, you can freely download the article without payment.
- When you get one relevant article, you can always get other relevant articles through the Cited by (see the yellow highlighted part in the Figure 1)

Part 2: Algorithm Analysis and Experiment

a. Algorithm Implementation & Experimentation.

- Select **AT LEAST THREE** algorithms for experimentation.
- **Choose one programming language** (either C++ or Python) and use it consistently for all implementations.
- You may use existing code from the internet, but proper citation is **mandatory** to avoid mark deductions.

b. Theoretical Analysis

- Theoretical analysis: analyse the design of each selected algorithm by using asymptotic analysis (show the steps).

c. Experimental analysis:

- analyse the execution time of your selected algorithms through experiments.
- You need to provide input with different number of input sizes (enough for you to construct the graph).
- Plot a graph which shows the execution time (y-axis, time could be in minutes/seconds/milliseconds) vs sizes of input (x-axis) of each algorithm.
- Provide a **detailed analysis** of the experimental results.

d. Comparison & Findings

- Compare theoretical and experimental results in terms of the growth of function and execution time.
- Identify discrepancies and explain possible reasons.

e. Proposal for Algorithm Improvement

Propose a new version that improve the algorithm or how you can modify the algorithm to solve other real-world problem (solution that is not yet discover by the other).

- Explain about your idea of design or solution
- Justify the possibility of implementing it

Plagiarism Policy

- **Strictly No Plagiarism:** Any form of plagiarism, including copying from previous works (seniors'/friends' reports, past projects, or online sources without proper citation), will result in an immediate **zero (0%) grade** for the project.
- **Proper Citations Required:** If using code or concepts from external sources, you must properly cite them.
- **Original Work Expected:** All analyses, discussions, and explanations must be in your own words.
- **Kindly note that the final report will be checked using Turnitin. Please ensure that the similarity index is below 15% and AI-generated content is below 20%.**

4) Report Formatting

- Font: Times New Roman
- Line Spacing: 1.5
- Font Size: Headers: Size 14, Content: Size 12
- Alignment: Justified paragraphs
- Tables & Figures: Must be properly labelled
- Reference format: Use the **IEEE referencing format** as outlined here: [VU IEEE Referencing Guide](https://libraryguides.vu.edu.au/ieeereferencing/gettingstarted).
(<https://libraryguides.vu.edu.au/ieeereferencing/gettingstarted>)
- Cover page: refer Appendix 1.

5) Submission Details

Part 1 Submission

- **Deadline:** 18/5/2025 by 5pm(Week 8).
- Format: PDF
- File Naming: LabSection_GroupNumber (e.g., 1A_Group1.pdf)
- Submission Platform: Microsoft Forms
- Required Contents in the report:
 - Cover Page
 - Table of Contents
 - Abstract
 - Introduction & Research Questions
 - Research Methodology
 - Related Studies
 - References

Part 2 Submission

- **Deadline:** 25/6/2025 by 5pm(Week 13).
- Format: ZIP (submitted as a .pdf file)
- File Naming: LabSection_GroupNumber (e.g., 1A_Group1.pdf)
- Submission Platform: Microsoft Forms
(Before submission, change the .zip file extension to .pdf as Microsoft Forms does not support ZIP files. The instructor will convert it back to ZIP after submission.)
- Required Files (inside ZIP):
 - [1] Group Declaration Form (No inactive members; any complaints from the leader will result in a zero mark for the reported member.)
 - [2] Final Report (Extended from Part 1, must include:)
 - Theoretical Analysis

- Experimental Analysis
 - Discussion/Suggestions/New Improvement Ideas
 - Conclusion
 - Appendix (Code for all algorithms)
- [3] Presentation Slides (*Refer to Appendix 3b*)
- [4] Program Files (*C++ or Python, properly named*)
- [5] Marking Scheme (Excel File)

***Any late submissions will NOT be accepted.**

Refer Rubrics.xlsx for the marking scheme.

6) Presentation

- The presentation will take place in week 14.
- Each group will have **10 minutes** to present their work, followed by a **5-minutes Q&A session**.
- Presentation Slide: refers to Appendix 3b

Final Submission Checklist

1. Final Report a. Cover page b. Table of Contents c. Abstract d. Introduction & Research Questions e. Research Method f. Related Studies g. Theoretical analysis h. Experimental analysis i. Discussion/Suggestion/New ideas of improvement j. Conclusion k. Appendix: Coding of those algorithms l. References	
2. Group declaration form	
3. Programs (.cpp & .py) ~	
4. Presentation slides (refers to Appendix 3b)	
5. Marking scheme (using excel file)	
6. Named the ZIP folder as: LabSection_GroupNo (e.g., 1A_Group1)	

